TRIBHUVAN UNIVERSITY — FACULTY OF MANAGEMENT

BBA Eighth Semester — BNK 209: Insurance and Risk Management (48 LHs)
COMPREHENSIVE EXAM REVISION NOTES, FIGURES, ACTUARIAL NUMERICALS & QUESTION BANK

Unit / Section	Official TU Syllabus Topics Covered in Full Detail	Lecture Hours
Unit 1: Introduction to Risk	Meaning of risk vs. uncertainty; Objective vs. subjective risk; Peril vs. Hazard (Physical, Moral, Attitudinal/Morale, Legal); Classification of risk (Pure vs. Speculative, Diversifiable vs. Non-Diversifiable, Enterprise vs. Systemic [Figure 1.1 & table]); Personal risks; Commercial risks; Burden of risk on society.	5 LHs
Unit 2: Risk Management	Meaning; Pre-loss & Post-loss objectives; 6-Step Risk Management Process (Figure 2.1); 2x2 Frequency-Severity Matrix (Figure 2.2); Risk Control (Avoidance, Prevention, Reduction) vs. Risk Financing (Retention/Captives, Non-Insurance Transfer, Commercial Insurance); Loss Forecasting (Probability, Regression, Distributions); NPV analysis; RMIS, VaR, Risk Appetite vs. Tolerance, TRM vs. ERM.	6 LHs
Unit 3: The Insurance Device	Definition & 4 basic characteristics of insurance; 6 requirements of an ideally insurable risk (and why Catastrophic Losses challenge insurance); Law of Large Numbers in risk pooling; Adverse selection & 5 control tools; Insurance vs. Gambling & Hedging; Benefits and costs of insurance to society.	5 LHs
Unit 4: Legal Principles in Insurance	The 6 Fundamental Legal Principles (Figure 4.1: Utmost Good Faith [Representations, Concealment, Warranties], Insurable Interest, Indemnity [ACV & exceptions], Subrogation, Contribution, Proximate Cause); 4 legal contract requirements; 5 distinct legal characteristics; Law & the Insurance Agent; 4 basic policy parts (D-I-C-E); Endorsements vs. Riders; Deductibles; Coinsurance formula & worked examples.	5 LHs
Unit 5: Life Insurance	Meaning & importance; Premature death & 6 financial impacts; Amount of life insurance to own (Human Life Value [HLV] vs. Financial Needs Approach); Types of life insurance (Figure 5.1 & Term vs. Cash-Value table); Contractual provisions; Dividend, Nonforfeiture & Settlement options; Riders (CI, PTD/AD&D;, Waiver); Nepal Income Tax Act rules; Actuarial life premium calculation.	10 LHs
Unit 6: Non-Life & Social Insurance	Life vs. Non-Life table; Commercial property & liability; Fire insurance ('Hostile Fire', policy types, rating factors & 4 rating methods: Judgment, Class, Schedule, Experience); Marine insurance (Hull, Cargo, General Average, LC role); Auto insurance & compulsory Third-Party Liability; Agriculture insurance in Nepal (80% subsidy); Social insurance (Nepal SSF 31% scheme).	12 LHs
Unit 7: Government Regulation	5 reasons for insurance regulation; 3 methods of regulation; 7 areas of regulation; Insurance Act 2079 & role/functions of Nepal Insurance Authority (NIA / Beema Pradhikaran); Role of insurance in Nepal's financial development.	5 LHs
Section 8: Numericals, Q&As; & Case Guide	Worked Actuarial Numericals (HLV, Needs Approach, 1-Yr Term NSP, Coinsurance, Contribution) + Top 25 Group A (2-Mark) Q&As; + Group D (20-Mark) Case Study Playbook.	Exam Bonus

UNIT 1: INTRODUCTION TO RISK & LOSS EXPOSURES (5 Lecture Hours)

1.1 Meaning of Risk, Uncertainty, Objective vs. Subjective Risk, Peril & Hazard

1. Meaning of Risk and Uncertainty:

In insurance and risk management (George E. Rejda), **Risk** is defined as *uncertainty concerning the occurrence of a financial loss* (or the variation between actual outcomes and expected outcomes in a situation where probabilities can be estimated).

• **Risk vs. Uncertainty (2-Mark / 5-Mark Distinction):**

Basis of Distinction	Risk	Uncertainty
1. Measurability	**Objectively Measurable:** Probabilities of possible outcomes can be calculated from historical data or actuarial tables.	**Not Objectively Measurable:** Probabilities cannot be quantified due to lack of historical precedent.
2. Nature	Objective phenomenon existing in the external world.	Subjective state of mind (personal doubt or mental anxiety).
3. Insurability	Can be pooled and transferred through commercial insurance.	Generally cannot be insured through standard actuarial policies.
4. Example	Probability of fire in 10,000 houses or mortality rate at age 35.	Whether a brand-new revolutionary product will succeed in 2035.

2. Objective Risk vs. Subjective Risk:

• **Objective Risk (Degree of Risk):** The relative variation of actual loss from expected loss:
Objective Risk = Standard Deviation of Losses / Expected Loss.
By the **Law of Large Numbers**, as the number of independent exposure units (N) increases, Objective Risk decreases inversely with the square root of N (1 / sqrt(N)). This is why insurance companies strive to build large pools of policyholders.

• **Subjective (Perceived) Risk:** Uncertainty based on an individual's mental condition or state of mind. Two individuals facing the exact same objective risk (e.g., a 1% chance of an accident) may perceive it differently depending on their personal risk aversion.

3. Peril vs. Hazard (And the Four Types of Hazards — Exam Essential):

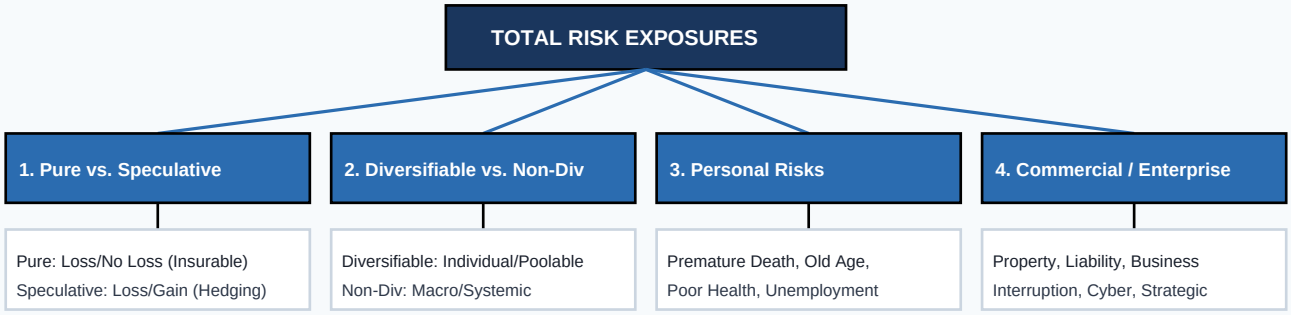
• **Peril:** The direct, immediate, and specific **cause of a loss** (e.g., fire, lightning, windstorm, flood, earthquake, theft, auto collision, premature death, sickness).

• **Hazard:** Any condition or situation that **creates or increases the frequency (probability) or severity (size)** of a loss arising from a given peril. There are four distinct types of hazards:

1. **Physical Hazard:** A tangible physical condition of the property, person, or environment that increases the chance of loss. *Examples: Icy or unpaved mountain roads (increases auto collision peril); faulty electrical wiring or storing kerosene in a warehouse (increases fire peril); obesity or high blood pressure (increases premature death peril).*
2. **Moral Hazard:** Dishonesty or character defects in an individual that increase the frequency or severity of loss—where the insured intentionally causes a loss or exaggerates a claim to defraud the insurer. *Examples: Deliberately setting fire to an unprofitable factory (arson) to collect fire insurance; staging a fake burglary; inflating hospital or garage repair bills.*
3. **Attitudinal Hazard (Morale Hazard):** Carelessness or indifference to a loss simply because one knows the loss is covered by insurance. *Examples: Leaving car keys in an unlocked car, failing to lock warehouse doors, or visiting expensive hospitals for minor ailments because health insurance pays 100%.*
4. **Legal Hazard:** Characteristics of the legal system or regulatory environment that increase the frequency or severity of liability losses. *Examples: Sympathetic juries awarding huge punitive damages against corporations; consumer protection laws expanding strict liability.*

1.2 Classification of Risk (Pure vs. Speculative, Diversifiable vs. Non-Diversifiable, Enterprise vs. Systemic)

FIGURE 1.1: CLASSIFICATION OF RISK & LOSS EXPOSURES IN RISK MANAGEMENT



Classification Pair	First Category (Definition & Examples)	Second Category (Definition & Examples)
1. Pure Risk vs. Speculative Risk	**Pure Risk:** A situation in which there are only two possible outcomes—**Loss or No Loss** (zero chance of financial gain). Accidental, involuntary, and **insurable** via pooling. *Examples:* Fire, earthquake, flood, premature death, permanent disability, auto accident, server data breach.	**Speculative Risk:** A situation in which there are three possible outcomes—**Loss, No Loss (Breakeven), or Financial Gain/Profit**. Undertaken voluntarily and **not insurable** (handled via derivative hedging or portfolio diversification). *Examples:* Investing/trading in NEPSE stocks, launching a new startup venture, betting, foreign exchange speculation.
2. Diversifiable vs. Non-Diversifiable Risk	**Diversifiable (Particular / Nonsystematic) Risk:** Affects only individuals or small groups and is not correlated with the macro-economy. Can be reduced or eliminated through diversification and insurance pooling. *Examples:* House fire, car theft, single-factory machinery breakdown.	**Non-Diversifiable (Fundamental / Systemic) Risk:** Affects the entire economy or large numbers of persons/groups simultaneously; cannot be eliminated by simple diversification. *Examples:* Hyperinflation, mass cyclical unemployment, global pandemic, war, catastrophic earthquake.
3. Enterprise Risk vs. Systemic Risk	**Enterprise Risk:** Encompasses all major risks faced by a single business firm across its operations: Pure risk + Speculative financial risk (interest, FX, credit, commodity) + Operational risk (IT/supply chain failure) + Strategic & Reputational risk.	**Systemic Risk:** The risk of collapse of an entire financial system or entire market (as opposed to a single entity), where the failure of one major interconnected bank or institution triggers a chain-reaction domino collapse across the economy.

1.3 Major Personal Risks, Commercial Risks & Burden of Risk on Society

1. Major Personal Risks (Loss Exposures Affecting Individuals & Families):

• **(1) Risk of Premature Death:** Death of a family breadwinner with unfulfilled financial obligations (dependent spouse/children, unpaid home mortgage or auto loan, unfunded college education).

• **(2) Risk of Insufficient Income During Retirement (Old-Age / Longevity Risk):** Outliving one's accumulated savings upon retirement, leading to a sharp drop in standard of living.

• **(3) Risk of Poor Health (Medical Expense & Disability Risk):** Catastrophic hospital/surgical bills combined with loss of earned income during prolonged disability.

• **(4) Risk of Involuntary Unemployment:** Loss of earned income due to economic recessions, technological automation, or corporate downsizing.

• **(5) Personal Property & Liability Risks:** Damage to one's home/car or legal liability for injuring others.

2. Major Commercial Risks (Loss Exposures Affecting Business Firms):

• **(1) Property Risk:** Direct physical damage to plants, offices, high-end servers, laptops, furniture, and inventory from fire, lightning, earthquake, flood, or burglary.

• **(2) Loss of Business Income (Business Interruption / Consequential Loss Risk):** Loss of net profit plus continuing fixed expenses (rent, salaries, loan interest) while the business is shut down for repairs after a physical loss, plus extra expense of renting temporary facilities.

• **(3) Liability Risk (Third-Party Legal Liability):** Lawsuits for bodily injury, property damage, defective products, motor accidents, workplace injuries, or professional errors & omissions (E&O;).

• **(4) Cyber Security, Data Breach & Identity Theft Risk:** Loss or corruption of sensitive customer data due to hacker attacks, ransomware, or employee server misconfiguration, triggering massive regulatory fines and client lawsuits.
• **(5) Human Resource (Personnel) Risk:** Death, disability, illness, or turnover of key executives and technical specialists.

3. Burden of Risk on Society (5-Mark Short Answer Classic):
• **1. Larger Emergency Funds Required (Idle Capital):** Without insurance, households and businesses must hold huge precautionary cash reserves, diverting capital away from productive investment.
• **2. Loss of Needed Goods and Services:** Fear of catastrophic liability or property loss discourages companies from producing vital high-risk products (e.g., vaccines, aviation, infrastructure).
• **3. Mental Worry, Fear, and Anxiety:** Psychological stress over potential disasters reduces human happiness and workplace focus.
• **4. Direct & Indirect Social Costs of Uncompensated Losses:** Uninsured business failures cause unemployment, family poverty, loan defaults, and higher government welfare burdens.

UNIT 2: RISK MANAGEMENT PROCESS & QUANTITATIVE TOOLS (6 Lecture Hours)

2.1 Concept, Objectives, Benefits & 6-Step Process of Risk Management

FIG 2.1: 6-STEP RISK MANAGEMENT PROCESS



FIG 2.2: FREQUENCY vs. SEVERITY RISK MATRIX (4 QUADRANTS)

Q1: LOW Freq + LOW Sev **Action: RETENTION** (Self-insure / Absorb)	**Q2: HIGH Freq + LOW Sev** **Action: LOSS PREVENTION** + Retention / Deductible
Q3: LOW Freq + HIGH Sev **Action: COMMERCIAL INSURANCE (Transfer!)**	**Q4: HIGH Freq + HIGH Sev** **Action: AVOIDANCE** (Abandon Activity!)

1. Pre-Loss and Post-Loss Objectives of Risk Management:

• Pre-Loss Objectives (Before a Loss Occurs):

1. *Economy:* Prepare for potential losses in the most cost-effective way (minimizing safety costs, insurance premiums, and uncompensated losses).
2. *Reduction of Anxiety:* Give peace of mind to managers, owners, and families.
3. *Meeting Legal & Contractual Obligations:* Comply with mandatory laws (third-party motor insurance, SSF/workers' compensation, safety codes) and bank loan covenants.

• Post-Loss Objectives (After a Catastrophic Loss Occurs):

1. *Survival of the Firm:* The most important post-loss objective—being able to resume operations after a major disaster.
2. *Continuity of Operations:* Resuming partial or full service quickly so customers do not defect to competitors.
3. *Stability of Earnings:* Maintaining steady EPS and cash flows without violent post-loss deficits.
4. *Continued Growth of the Firm:* Preserving capital for R&D; and market expansion.
5. *Social Responsibility:* Minimizing the adverse impact of a disaster on employees, suppliers, customers, and the community.

2. Detailed 6-Step Process of Risk Management (10-Mark Exam Classic):

• Step 1: Determination of Objectives: Establish formal pre-loss and post-loss objectives in a written corporate Risk Management Policy Statement.

• Step 2: Identification of Loss Exposures: Systematically uncover all pure and enterprise loss exposures (Property, Liability, Business Interruption, Human Resources, Crime/Cyber) using five identification tools: (i) *Risk Analysis Questionnaires/Checklists*, (ii) *Physical Plant Inspections*, (iii) *Flowcharts of Production & Delivery*, (iv) *Financial Statement Analysis*, and (v) *Historical Loss/Claims Records*.

• Step 3: Measurement and Evaluation of Loss Exposures (Loss Forecasting): Estimate for each exposure:

- **Loss Frequency:** Probable number of losses during the budget period.
- **Loss Severity:** Probable size of losses—measuring both **Maximum Possible Loss** (worst-case total catastrophe) and **Probable Maximum Loss (PML)** (worst loss likely to happen under normal protection). Severity is more critical than frequency because a single catastrophic loss can wipe out a firm!

• Step 4: Selection of Appropriate Risk Treatment Techniques: Choose the optimal combination of **Risk Control** (Avoidance, Loss Prevention, Loss Reduction [Duplication & Separation]) and **Risk Financing** (Retention/Self-Insurance/Captives, Non-Insurance Transfer, Commercial Insurance) based on the 2x2 Frequency-Severity Matrix (Figure 2.2).

• Step 5: Implementation of the Risk Management Program: Purchase selected insurance policies, set up retention reserves, install fire/cyber safety systems, and train employees across all departments.

• Step 6: Regular Monitoring, Review, and Evaluation: Periodically audit claims, safety compliance, and policy limits using a **Risk Management Information System (RMIS)**.

2.2 Loss Forecasting, Financial Analysis in Risk Management, RMIS, VaR & ERM

1. Loss Forecasting as a Fundamental Activity in Risk Management (10-Mark / 5-Mark Question):
Loss forecasting is the quantitative prediction of future loss frequency and severity so the risk manager can budget retention funds, evaluate insurance quotes, and perform cost-benefit analysis of safety equipment. The three primary techniques of loss forecasting are:
• **(i) Probability Analysis:** Uses historical loss frequency and severity probabilities to calculate the **Expected Loss [E(L) = Sum (Loss_i x Prob_i)]** and standard deviation for independent and dependent events.
• **(ii) Regression Analysis:** Establishes a statistical relationship between a dependent loss variable *Y* (e.g., annual number of workers' compensation claims or auto accidents) and an independent physical exposure variable *X* (e.g., total payroll hours or fleet kilometers driven): **Y = a + bX**. When next year's budgeted payroll or mileage *X* is plugged in, regression forecasts expected losses *Y*.
• **(iii) Forecasting Based on Theoretical Loss Distributions:** Models loss frequency using the **Poisson or Binomial distribution** and loss severity using the **Lognormal or Pareto distribution** (often via Monte Carlo simulation) to estimate tail risk and Probable Maximum Loss.

2. Financial Analysis in Risk Management Decision Making (Time Value of Money / NPV):
Because safety investments (e.g., installing an automatic fire sprinkler system or backup server) require an immediate cash outflow today while generating cash inflows (reduced uninsured losses and lower fire insurance premiums) over many future years, risk managers use **Net Present Value (NPV)** and **Internal Rate of Return (IRR)** to evaluate risk control projects.

3. Risk Appetite vs. Risk Tolerance, Value at Risk (VaR), RMIS & Enterprise Risk Management (ERM):
• **Risk Appetite vs. Risk Tolerance:** *Risk Appetite* is the broad, qualitative strategic level of risk an organization is willing to accept in pursuit of value; *Risk Tolerance* is the specific, quantitative maximum acceptable deviation around a particular objective.
• **Value at Risk (VaR):** A statistical measure that summarizes the **worst expected financial loss** over a target time horizon (e.g., 1 day, 10 days, or 1 year) at a given confidence level (e.g., 95% or 99%) under normal market conditions.
• **Risk Management Information System (RMIS):** A computerized database that stores property values, policy schedules, premiums, and claims histories, enabling automated loss forecasting, trend detection, and renewal tracking.
• **Traditional Risk Management (TRM) vs. Enterprise Risk Management (ERM):** TRM manages only *insurable pure risks* in fragmented departmental silos; ERM holistically manages *all risks* (Pure, Speculative Financial, Operational, Cyber, Strategic) across the entire enterprise under a **Chief Risk Officer (CRO)**, using natural hedges, derivatives, and insurance together.

UNIT 3: THE INSURANCE DEVICE & RISK POOLING (5 Lecture Hours)

3.1 Definition & Basic Characteristics of Insurance, Law of Large Numbers & Insurable Risk

1. Four Basic Characteristics of Insurance:
• **(1) Pooling of Losses (Risk Pooling / Loss Sharing):** Spreading losses incurred by the few over the entire group, so that in the process, **average loss is substituted for actual loss**.
• **(2) Payment of Fortuitous Losses:** A fortuitous loss is one that is unforeseen, unexpected, and occurs as a result of chance (accidental, not intentional).
• **(3) Risk Transfer:** A pure risk is transferred from the insured to the insurer, who is in a stronger financial position to pay the loss.
• **(4) Indemnification:** The insured is restored to their approximate pre-loss financial position after a loss occurs.

2. Six Requirements of an Ideally Insurable Risk (And Why Catastrophic Losses Pose a Challenge):
• **1. Large Number of Exposure Units:** Needed so the Law of Large Numbers can predict average loss accurately.
• **2. Accidental and Unintentional Loss:** Intentional losses create moral hazard and violate actuarial randomness.
• **3. Determinable and Measurable Loss:** Cause, time, place, and dollar amount must be verifiable.
• **4. No Catastrophic Loss:** Ideally, a large proportion of exposure units should not suffer loss at the same time.
(Why Catastrophes Challenge Insurance: Earthquakes, widespread floods, wars, and pandemics hit thousands of insured units simultaneously—breaking the independence assumption of the Law of Large Numbers, threatening

insurer solvency, and requiring global reinsurance, catastrophe bonds, and government pools.)
• **5. Calculable Chance of Loss:** Average frequency and average severity must be estimable.
• **6. Economically Feasible Premium:** The insured must be able to afford the premium (chance of loss cannot be too high, e.g., >40%).

3.2 Adverse Selection, Insurance vs. Gambling, Insurance vs. Hedging & Benefits/Costs to Society

1. Adverse Selection & Tools to Control It (5-Mark Exam Favourite):
• **Definition:** Adverse selection is the tendency of persons with a higher-than-average probability of loss to seek insurance at standard (average) rates, which—if uncontrolled—results in higher-than-expected claims.
• **Five Tools Used by Insurers to Control Adverse Selection:**
1. *Careful Underwriting & Risk Classification:* Selecting and classifying applicants into standard, preferred, substandard, or declined categories based on age, medical history, occupation, and property inspections.
2. *Waiting Periods & Pre-Existing Condition Clauses:* Imposing waiting periods (e.g., 30 days to 2 years in health insurance) and suicide exclusions.
3. *Deductibles, Coinsurance & Policy Limits:* Requiring cost-sharing by the insured.
4. *Group Insurance Underwriting Rules:* Requiring automatic enrollment or high participation (e.g., 75%+) across an employer group.
5. *Enforcing Utmost Good Faith:* Rescinding policies where applicants conceal material health/risk facts.

2. Benefits and Costs of Insurance to Society:
• **Social & Economic Benefits of Insurance:** (1) Indemnification for losses (restores families and businesses), (2) Reduction of worry and fear, (3) Source of long-term investment funds (mobilizes savings for banks, infrastructure, and hydropower), (4) Loss prevention incentives (schedule rating discounts for fire sprinklers, workplace safety), and (5) Enhancement of credit (collateral protection for mortgages, auto loans, and import LCs).
• **Social Costs of Insurance:** (1) Cost of doing business (operating expenses and commissions—loading), (2) Fraudulent claims (moral hazard), and (3) Inflated claims (attitudinal hazard).

UNIT 4: LEGAL PRINCIPLES & CONTRACT PROVISIONS IN INSURANCE (5 Lecture Hours)

4.1 The Six Fundamental Legal Principles of Insurance Contracts (10-Mark Exam Essential)

FIGURE 4.1: THE SIX FUNDAMENTAL LEGAL PRINCIPLES OF INSURANCE CONTRACTS

1. Utmost Good Faith	2. Insurable Interest	3. Indemnity	4. Subrogation	5. Contribution	6. Proximate Cause
Uberrimae Fidei: • Representations • Concealment • Warranties	Pecuniary stake: • Life: At Inception • Property: At Loss • Stops Gambling	Restore pre-loss position only: • ACV = RC - Dep. • No profit on loss	Insurer steps into insured's shoes to sue negligent 3rd party after paying	Double insurance pro-rata sharing: Policy / Total Limits x Loss	Causa Proxima: Active, dominant efficient cause must be insured peril

In-Depth Explanation of the Six Fundamental Legal Principles:

• 1. Principle of Utmost Good Faith (Uberrimae Fidei):

- *Concept:* Unlike ordinary commercial sales (*caveat emptor*), an insurance contract requires the highest degree of honesty from both the applicant and the insurer. They must voluntarily disclose all material facts relating to the risk.

- *Three Supporting Legal Doctrines (5-Mark Question!):*

1. **Representations:** Statements made by the applicant on the proposal form before policy issuance. A contract is voidable if a representation is (a) *material*, (b) *false*, and (c) *relied upon by the insurer*.
2. **Concealment:** Intentional silence or failure of the applicant to disclose a known material fact. To void a policy, the insurer must prove the concealed fact was material and intentionally hidden.
3. **Warranties:** Statements or promises incorporated directly into the policy contract that guarantee a condition exists (*affirmative warranty*) or will be maintained throughout the policy term (*promissory warranty*—e.g., maintaining a burglar alarm or working fire sprinklers). Strict compliance is required.

• 2. Principle of Insurable Interest:

- *Concept:* The insured must stand in a legally recognized financial relationship to the subject matter of insurance such that they will suffer a direct pecuniary (financial) loss if the insured event occurs.

- *Three Purposes:* (i) To prevent gambling/wagering on strangers' lives or property, (ii) To reduce moral hazard (murder or arson for insurance money), and (iii) To measure the amount of the insured's loss in property insurance.

- *When Insurable Interest Must Exist:*

• In **Life Insurance**, insurable interest must exist **only at the inception of the policy** (not necessarily at the time of death).

• In **Property and Marine Insurance**, insurable interest must exist **at the time of the loss** (and generally at inception).

• 3. Principle of Indemnity:

- *Concept:* The insurer agrees to pay no more than the actual amount of the loss; stated differently, the insured should not profit from a loss, but should be restored to the same financial position that existed prior to the loss.

- *Methods of Determining Actual Cash Value (ACV):*

1. **Replacement Cost Minus Depreciation: ACV = Current Replacement Cost - Physical Depreciation**.
2. **Fair Market Value** (what a willing buyer would pay a willing seller before the loss).
3. **Broad Evidence Rule** (considering all relevant factors: replacement cost, market value, rental income, expert appraisal).

- *Four Exceptions to the Principle of Indemnity:* (1) **Valued Policies** (pays full face value upon total loss, e.g., fine art or marine hull), (2) **Valued Policy Laws**, (3) **Replacement Cost Insurance** (pays new replacement cost without deducting depreciation), and (4) **Life Insurance** (a valued benefit contract, since human life cannot be measured by replacement cost minus depreciation).

• 4. Principle of Subrogation (Corollary of Indemnity):

- *Concept:* Substitution of the insurer in place of the insured for the purpose of claiming indemnity from a negligent third party for a loss covered by insurance. Once the insurer pays the insured's claim, the insurer steps into the legal shoes of the insured and can sue the negligent tortfeasor to recover the amount paid.
- *Key Rules:* (i) Applies only to property and liability contracts of indemnity (**never applies to Life Insurance**), (ii) The insurer is entitled to recover only up to the amount it paid out to the insured, and (iii) The insured cannot impair the insurer's subrogation rights by releasing the negligent party.

• **5. Principle of Contribution (Other-Insurance Provision / Corollary of Indemnity):**

- *Concept:* Applies when two or more property/liability policies cover the **same insurable interest, same subject matter, and same peril** at the same time (Double Insurance). To prevent the insured from collecting more than 100% of the actual loss, each insurer contributes proportionately:
- **Pro-Rata Contribution Formula:**

Insurer A's Share = Actual Loss x [Policy A Limit / Sum of All Policies' Limits].

• **6. Principle of Proximate Cause (Causa Proxima):**

- *Concept:* Proximate cause is the active, efficient cause that sets in motion an unbroken chain of events which brings about a result, without the intervention of any new and independent force. For a claim to be payable, the proximate cause of the loss must be an **insured peril**.

4.2 Requirements, Distinct Legal Characteristics, Agency Law, Basic Parts & Policy Provisions

1. Four Legal Requirements of a Valid Insurance Contract:
(1) **Offer and Acceptance** (applicant submits proposal + first premium; insurer accepts/issues binder), (2) **Consideration** (insured pays premium + complies with conditions; insurer promises to indemnify), (3) **Competent Parties** (legal age, sound mind, licensed insurer), and (4) **Legal Purpose** (must have insurable interest; not a wagering contract).

2. Five Distinct Legal Characteristics of Insurance Contracts:
(1) **Aleatory Contract** (unequal dollar exchange depending on an uncertain event), (2) **Unilateral Contract** (only the insurer makes an enforceable legal promise after premium payment), (3) **Conditional Contract** (insurer's duty to pay depends on fulfilling policy conditions), (4) **Personal Contract** (in property insurance, insures the owner—cannot be assigned without insurer consent), and (5) **Contract of Adhesion** (drafted by the insurer; any ambiguity is construed in favor of the insured).

3. Law and the Insurance Agent (Role of Insurance Agent):
An insurance agent is the legal representative of the principal (insurer) acting under Express, Implied, and Apparent authority. Under agency law: (i) Knowledge of the agent is imputed to the insurer, (ii) The agent performs field underwriting ("eyes and ears of the insurer"), educates clients, collects premiums, services policies, and assists families during claim settlement.

4. Basic Parts of an Insurance Policy (D-I-C-E) & Key Provisions (Endorsements, Deductibles, Coinsurance):
• **Four Basic Parts: (1) Declarations** (identity of insured, property description, policy period, sum insured, premium), **(2) Insuring Agreement** (*Named-Perils* vs. *Open-Perils / All-Risk* coverage), **(3) Exclusions**, and **(4) Conditions**.
• **Endorsements vs. Riders:** Written provisions that add, delete, or modify coverage in the standard policy (called **Endorsements** in Property/Casualty insurance and **Riders** in Life/Health insurance).
• **Deductibles (Straight, Aggregate, Waiting-Period):** Provision by which a specified amount is subtracted from the loss payment otherwise payable to the insured—eliminates small administrative claims, lowers premiums, and reduces moral/attitudinal hazard.
• **Coinsurance (Condition of Average in Commercial Property Insurance) — Numerical Formula:**
Requires the insured to insure property for at least a stated percentage (e.g., 80% or 100%) of its true insurable value at the time of loss:
Amount of Recovery = Actual Loss x [Amount of Insurance Carried / Amount of Insurance Required] (never exceeding the policy face limit or actual loss).

PRACTICAL / NEPAL CORPORATE EXAMPLE:

Worked Numerical Example on Coinsurance & Contribution:

• **Coinsurance Example:** A commercial building has a Replacement Cost (True Value) of Rs 10,000,000 and an 80% coinsurance clause (so Insurance Required = 80% x 10M = Rs 8,000,000). The owner carries only Rs 6,000,000 of fire insurance. If a fire causes a Rs 2,000,000 loss:

Claim Paid = Rs 2,000,000 x (6,000,000 / 8,000,000) = Rs 1,500,000 (the owner bears the remaining Rs 500,000 as a coinsurance penalty!).

• **Pro-Rata Contribution Example:** A warehouse is insured against fire with Insurer A for Rs 3,000,000 and Insurer B for Rs 2,000,000 (Total Insurance = Rs 5,000,000). A fire causes an actual loss of Rs 1,000,000:

- **Insurer A pays:** 1,000,000 x (3,000,000 / 5,000,000) = **Rs 600,000**
- **Insurer B pays:** 1,000,000 x (2,000,000 / 5,000,000) = **Rs 400,000** (Total = Rs 1,000,000).

UNIT 5: LIFE INSURANCE & ACTUARIAL RATE MAKING (10 Lecture Hours)

5.1 Meaning, Importance, Premature Death & Financial Impact of Premature Death

1. Meaning and Importance of Life Insurance:
Life insurance is a legal contract between a policyholder and a life insurer under which the insurer promises to pay a designated lump-sum benefit (Sum Assured plus accrued bonuses, if any) to the named beneficiary upon the death of the insured life, or to the insured upon survival to the maturity of an endowment policy, in exchange for periodic premium payments.
• **Importance:** (1) Immediate estate creation for young families, (2) Protection against loss of Human Life Value, (3) Mortgage and business debt redemption, (4) Disciplined long-term tax-advantaged savings for retirement and children's education, and (5) Mobilization of domestic capital for national economic development.

2. Premature Death and Its Financial Implications (5-Mark Exam Classic):
• **Definition:** Premature death is the death of a family head (breadwinner) who leaves behind unfulfilled financial obligations—such as dependent spouse, children, or aging parents to support, an unpaid home mortgage or auto loan, or unfunded children's college education.
• **Six Financial Implications (Costs) of Premature Death:**
1. *Permanent Loss of Human Life Value:* The family loses its share of the deceased breadwinner's future net earning power.
2. *Additional Immediate Cash Outflows (Clearance / Final Expenses):* High funeral/cremation expenses, uninsured final medical/ICU bills, estate settlement fees, and unpaid taxes.
3. *Debt and Mortgage Default Risk:* Inability to service home loans, vehicle loans, or business loans, risking bank foreclosure of the family home.
4. *Reduction in Standard of Living:* Because ongoing family income falls below family financial needs, surviving dependents suffer a permanent drop in living standards.
5. *Disruption of Children's Higher Education Plans:* Lack of funds for school/university tuition.
6. *Noneconomic Costs:* Intense emotional grief and loss of parental guidance and role modeling.

5.2 Determining the Amount of Life Insurance to Own: Human Life Value (HLV) vs. Needs Approach

1. Method A — The Human Life Value (HLV) Approach (Solomon S. Huebner):
Defines Human Life Value as the **present value of the family's share of the deceased breadwinner's future net earnings** over their remaining working career:
• **Step 1:** Calculate Remaining Working Years: **n = Planned Retirement Age - Current Age**.
• **Step 2:** Estimate Average Annual Gross Earnings over the remaining working career.
• **Step 3:** Deduct the breadwinner's personal self-maintenance expenses, income taxes, and personal life/health insurance premiums to find the **Net Annual Family Share**.
• **Step 4:** Discount the stream of annual family contributions to Present Value at a reasonable discount rate (i):
Human Life Value (HLV) = Net Annual Family Share x PVIFA(i%, n years)
where **PVIFA(i, n) = [1 - (1 + i)^(-n)] / i**.

2. Method B — The Financial Needs Approach:
Estimates the total life insurance required by summing the specific financial needs of the surviving family and subtracting existing liquid financial assets:
• **(1) Estate Clearance Fund (Immediate Cash Needs):** Funeral expenses, uninsured medical bills, probate costs, and short-term installment debts.
• **(2) Readjustment Period Income (1–2 Years):** Temporary income to let the family adjust smoothly without drastic lifestyle cuts.
• **(3) Dependency Period Income:** Present value of annual living expenses needed until the youngest child reaches financial independence (age 18–22) or for the surviving spouse's/parents' lifetime.
• **(4) Mortgage / Bank Loan Redemption Fund:** Lump sum to pay off the outstanding home loan and vehicle loan

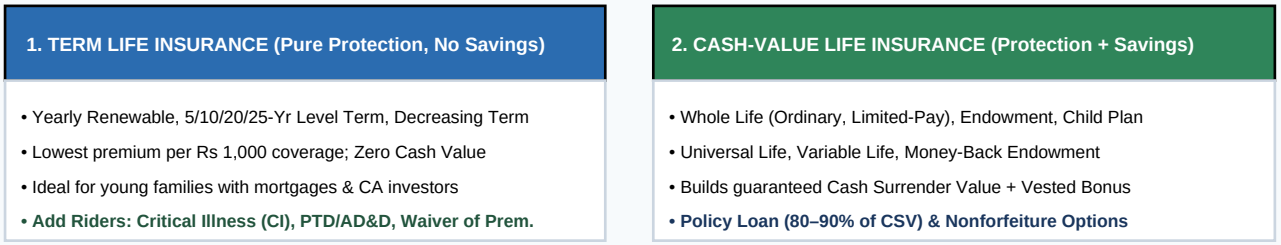
immediately.

• **(5) Special Funds (Education & Emergency Funds):** Present value (or nominal corpus) for children's university education plus a family emergency reserve.

• **Formula: New Life Insurance Needed = Total Financial Needs - Existing Liquid Assets & Existing Life Insurance**.

5.3 Types of Life Insurance Policies: Term Life vs. Cash-Value Life Insurance (10-Mark Favourite)

FIGURE 5.1: TAXONOMY OF LIFE INSURANCE POLICIES & ACTUARIAL RATE MAKING



Basis of Comparison	Term Life Insurance	Cash-Value Life Insurance (Whole Life / Endowment)
1. Period of Coverage	**Temporary:** Covers a fixed term (1, 5, 10, 20, or 30 years) and expires without value if the insured survives.	**Permanent / Long-Term:** Covers the entire lifetime (Whole Life) or pays out at a fixed maturity date (Endowment).
2. Savings / Cash Value	**Pure Protection Only (Zero Cash Value):** Has no investment or savings accumulation.	**Protection + Savings:** Builds a guaranteed **Cash Surrender Value (CSV)** plus vested reversionary bonuses.
3. Premium Level & Cost	**Lowest initial premium** per Rs 1,000 of death benefit; increases at each renewal as age rises.	**Much higher level premium** (3x to 8x higher initially) because excess early premiums build the cash reserve.
4. Maturity / Survival Benefit	Pays **only if death occurs** during the policy term (nothing paid on survival).	Pays the full Sum Assured + Bonus on death OR upon surviving to maturity (Endowment) / age 100 (Whole Life).
5. Policy Loans & Surrender	Cannot be surrendered for cash and has no loan value.	Policyholder can surrender for cash or take a policy loan up to 80%–90% of accumulated surrender value.
6. Best Suitability	Young breadwinners with high mortgages/children and limited cash flow, or financially savvy investors who **'Buy Term and Invest the Difference'**.	Individuals wanting forced long-term savings, guaranteed child education/retirement corpus, or permanent estate liquidity.

• **Major Sub-Types of Life Insurance Policies:**

1. **Term Life Variants:** (i) *Yearly Renewable Term (YRT)*, (ii) *5/10/20/25-Year Level Term*, (iii) *Decreasing Term (Mortgage Protection)*, (iv) *Convertible Term* (can be converted into Whole Life/Endowment without medical exam), and (v) *Return-of-Premium Term*.
2. **Whole Life Variants:** (i) *Ordinary (Straight) Whole Life* (level premiums paid until death/age 100), (ii) *Limited-Payment Whole Life* (coverage for whole life, but premiums paid for only 10, 15, or 20 years), and (iii) *Single-Premium Whole Life*.
3. **Endowment & Money-Back Policies (Most Popular in Nepal):** Pays Sum Assured + Bonus either on death within the term or upon survival at the end of the term (Money-Back policies pay periodic survival installments every 4–5 years plus full death cover).
4. **Universal Life, Variable Life & Group Life Insurance:** *Universal Life* unbundles protection and savings with flexible premiums; *Variable Life* invests cash value in mutual-fund subaccounts; *Group Life Insurance* covers all employees of a company under one master contract.

5.4 Life Insurance Contractual Provisions, Options, Riders, Taxation & Premium Calculation

1. Standard Life Insurance Contractual Clauses:

• **Incontestable Clause:** After the policy has been in force for a specified period (typically 2 years during the insured's lifetime), the insurer cannot contest or void the death claim on grounds of unintentional misrepresentation.

• **Suicide Clause:** If the insured commits suicide within the first 1–2 years of policy issue, the insurer does not pay the face value (only refunds premiums paid).

• **Grace Period Clause:** Grants a 30- or 31-day grace period after a renewal premium due date during which coverage stays in force.

• **Reinstatement (Revival) Clause:** Allows a lapsed policy to be revived within 2–5 years by paying overdue premiums with interest and providing evidence of insurability.

• **Misstatement of Age or Sex Clause:** Adjusts the benefit amount to what the premium paid would have purchased at the true age/sex.

• **Beneficiary Designation (Primary vs. Contingent; Revocable vs. Irrevocable).**

2. Nonforfeiture Options, Dividend (Bonus) Options & Settlement Options:

• **Three Nonforfeiture Options (If a Cash-Value Policyholder Stops Paying Premiums):** (1) *Cash Surrender Value* (take cash and terminate policy), (2) *Reduced Paid-Up Insurance* (keep permanent policy for a smaller face amount with no further premiums), and (3) *Extended Term Insurance* (keep full face value for a limited term).

• **Four Dividend (Participating Policy) Options:** (1) Take cash, (2) Reduce next premium, (3) Leave with insurer to accumulate at interest, or (4) Buy paid-up additions (reversionary bonus).

• **Five Settlement Options (How Death/Maturity Proceeds Are Paid):** (1) Lump-sum cash, (2) Interest option, (3) Fixed-period option, (4) Fixed-amount option, and (5) Life income (annuity) option.

3. Additional Life Insurance Benefits (Riders — Very High Exam Frequency!):

• **(1) Waiver of Premium Disability Rider:** If the insured becomes totally and permanently disabled before age 60/65, all future premiums are waived while the policy remains 100% in force.

• **(2) Accidental Death and Dismemberment (AD&D; / Double Indemnity) & Permanent Total Disability (PTD) Rider:** Doubles the death benefit if death results from an accident, or pays monthly income/lump sum upon permanent disability.

• **(3) Critical Illness (CI) Rider (Accelerated Death Benefit):** Pays an immediate tax-free lump sum upon first diagnosis of specified dread diseases (cancer, heart attack, stroke, kidney failure, coronary bypass, organ transplant) to fund medical treatment and lost income.

• **(4) Guaranteed Purchase Option (Insured Insurability Rider):** Allows young insureds to buy additional life insurance at future milestones without a new medical exam.

4. Taxation of Life Insurance in Nepal (Income Tax Act 2058):

• **Tax Benefit on Premiums Paid:** A resident natural person can deduct annual life insurance premiums paid up to **Rs 40,000** from their taxable income (in addition to up to Rs 20,000 for health insurance and Rs 5,000 for personal house insurance).

• **Tax on Payouts:** Death benefit payouts to beneficiaries are **100% exempt from income tax**. For maturity/investment gains on policies issued by resident insurers, a **5% final withholding tax** applies on the net gain (Maturity Payout minus Total Premiums Paid).

5. Actuarial Process of Calculating Life Insurance Premiums (5-Mark Question + Numerical):

Life insurance premiums are calculated using the **Actuarial Equivalence Principle** (Present Value of Expected Premiums = Present Value of Expected Benefits) based on three factors: **(1) Mortality Rate (q_x)** from the Mortality Table, **(2) Assumed Interest Rate (i)**, and **(3) Loading** for expenses/contingencies:

• **Formula for 1-Year Term Net Single Premium (NSP):**

NSP = [Policy Face Value x Probability of Death at Age x (q_x)] / (1 + i)^1

• **Formula for Gross (Office) Premium:**

Gross Premium = Net Premium + Loading for Expenses, Commissions, Taxes & Bonus.

UNIT 6: NON-LIFE (GENERAL) INSURANCE & SOCIAL INSURANCE (12 Lecture Hours)

6.1 Commercial Property & Liability Insurance and Fire Insurance (Nature, Policies, Rating & Claims)

1. Life Insurance vs. Non-Life (General) Insurance (Comparison Table):

Basis of Difference	Life Insurance	Non-Life (General) Insurance
1. Subject Matter	Human life (health, longevity, mortality).	Property, physical assets, cargo, motor vehicles, and third-party legal liability.
2. Nature of Contract	**Valued (Benefit) Contract:** Pays the agreed Sum Assured + Bonus; not a contract of strict indemnity.	**Strict Contract of Indemnity:** Pays only the actual financial loss suffered (up to sum insured).
3. Certainty of Event	Death is certain (only the timing of death is uncertain).	Occurrence of the peril (fire, accident, flood) is uncertain and may never happen.
4. Policy Duration	Long-term contract (5, 10, 20, 30 years, or whole life).	Short-term contract (typically **1 year**, renewable annually).
5. Savings / Investment Element	Cash-value policies (Endowment/Whole Life) include a major savings/bonus element.	No savings or surrender value; pure risk protection only.
6. Insurable Interest Timing	Must exist at the **inception** of the policy.	Must exist at the **time of loss** (and inception).

2. Fire Insurance: Nature, Types of Policies, Rate Fixation & Claim Settlement (10-Mark Essential):

• **Legal Meaning of "Fire":** Under a Standard Fire Policy, "fire" requires: (1) *Actual ignition/flame* (mere heating, scorching, or fermentation is excluded), and (2) *Accidental "Hostile Fire"* that escapes its intended receptacle (as opposed to a *friendly fire* inside a stove/furnace).

• **Major Types of Fire Insurance Policies:**

1. *Specific Policy:* Insures property for a specified sum without average clause.
2. *Average Policy (Standard in Nepal):* Contains the Condition of Average (coinsurance clause) penalizing under-insurance proportionately.
3. *Valued Policy:* Agreed value fixed at inception (for artworks, antiques, jewelry).
4. *Floating (Floater) Policy:* Covers fluctuating inventory stored across multiple warehouses under a single sum insured.
5. *Declaration Policy:* Designed for businesses with seasonal stock fluctuations; insured declares monthly stock values and premium is adjusted at year-end.
6. *Reinstatement (Replacement Cost) Policy:* Pays the cost of replacing damaged machinery/buildings with new assets of the same kind without deducting depreciation.
7. *Consequential Loss (Loss of Profits / Business Interruption) Policy:* Covers lost net profit and standing charges during post-fire rebuilding.

• **Factors Influencing Rate Fixation in Fire Insurance:**

(1) *Construction Material & Design* (RCC Class A vs. timber/thatched), (2) *Occupancy / Nature of Goods* (office vs. chemical/garment factory), (3) *Location & Exposure Hazard* (proximity to hazardous neighbors), (4) *Internal Fire-Protection Appliances* (sprinklers, extinguishers, smoke alarms), (5) *Public Fire Brigade Access*, and (6) *Past Loss Experience & Housekeeping*.

• **Four Methods of Determining Rates in Fire Insurance:**

1. **Judgment Rating:** Underwriter sets a custom rate for a unique risk based on professional judgment.
2. **Class (Manual / Tariff) Rating:** Groups similar buildings (e.g., RCC residential homes) into homogeneous classes with a uniform manual rate.
3. **Schedule Rating:** Starts with a baseline "basis rate" for a standard building, **adds debits (surcharges)** for physical hazards (bad wiring, flammable storage), and **subtracts credits (discounts)** for fire sprinklers, alarms,

and watchmen.
4. **Experience Rating:** Modifies the rate based on the large commercial client's own historical claims ratio.

6.2 Marine Insurance, Auto (Motor) Insurance, Agriculture Insurance & Social Insurance

1. Marine Insurance (Nature, Policies & Role in International Trade):
• **Subject Matter:** (i) *Hull Insurance* (vessel/ship), (ii) *Cargo Insurance* (goods in ocean/air/inland transit under Institute Cargo Clauses A [All Risks], B, and C), (iii) *Freight Insurance*, and (iv) *Marine Liability*.
• **Types of Marine Policies:** *Voyage Policy* (from Port A to Port B), *Time Policy* (for 1 year), *Mixed Policy*, *Open Cover / Floating Cargo Policy* (for regular importers/exporters), and *Valued vs. Unvalued Policy*.
• **Marine Losses: Actual Total Loss**, **Constructive Total Loss** (cost of salvage/repair exceeds insured value, so insured abandons vessel to insurer), **Particular Average** (partial loss borne by the specific cargo owner), and **General Average** (voluntary sacrifice like jettisoning cargo in a storm to save the whole voyage, shared proportionately by all parties).
• **Role in International Trade:** Mandatory for opening bank Letters of Credit (LC), enables CIF/CIP Incoterms trade, and protects landlocked importers (like Nepalese businesses importing via Kolkata/Visakhapatnam ports).

2. Auto (Motor) Insurance & Compulsory Third-Party Liability Insurance:
• **Coverages in a Comprehensive Motor Policy:**
1. *Part A — Third-Party Liability Coverage (Compulsory by Law):* Covers bodily injury, permanent disability, death, and property damage caused by the insured vehicle (First Party) to innocent outsiders (Third Parties), plus legal defense costs.
2. *Part B — Medical Payments / Driver, Helper & Passenger Personal Accident Coverage.*
3. *Part C — Uninsured / Hit-and-Run Motorists Coverage.*
4. *Part D — Own Damage (OD) Coverage:* Covers repair/replacement of the insured's own vehicle due to collision, overturning, fire, theft, flood, landslide, or riot (RSMD).

3. Agriculture Insurance in Nepal & Social Insurance (SSF):
• **Agriculture Insurance:** Covers crops (paddy, maize, vegetables, fruits), livestock (cattle, buffaloes, goats), poultry, fish, and herbs against natural perils and epidemics. In Nepal, the government provides up to an **80% subsidy on agriculture insurance premiums**—vital because >60% of the population relies on farming, terrain is disaster-prone, and insured crops/cattle serve as bank loan collateral.
• **Social Insurance:** Compulsory government-mandated programs financed by earmarked payroll contributions to provide a floor of income security:
1. *Old-Age, Survivors, and Disability Insurance (OASDI / Pension)*
2. *Medicare / National Health Insurance*
3. *Unemployment Insurance*
4. *Workers' Compensation Insurance* (occupational injury/disease)
5. *Nepal's Contribution-Based Social Security Fund (SSF):* 31% monthly contribution (20% employer + 11% employee of basic salary) funding Medical/Maternity, Accident/Disability, Dependent Family, and Old-Age Pension schemes.

UNIT 7: GOVERNMENT REGULATION OF INSURANCE & NEPAL INSURANCE AUTHORITY (5 Lecture Hours)

7.1 Reasons for Insurance Regulation, Methods & Areas of Regulation

1. Five Core Reasons Why Insurance is Heavily Regulated (10-Mark Exam Favourite):
• **(1) Maintain Insurer Solvency (Protect Public Trust & Financial Stability):** Premiums are paid upfront while claims may arise decades later. If an insurer fails, widows, pensioners, and disaster victims lose their financial lifeline.
• **(2) Compensate for Inadequate Consumer Knowledge (Information Asymmetry):** Insurance policies are complex technical legal contracts of adhesion; ordinary citizens cannot judge policy exclusions or an insurer's financial reserves on their own.
• **(3) Ensure Reasonable, Adequate, and Non-Discriminatory Rates:**

- *Adequate:* High enough to pay claims and prevent cut-throat price wars that bankrupt insurers.
- *Not Excessive:* Prevent cartel price-gouging.
- *Not Unfairly Discriminatory:* Similar risks must pay equitable rates.

• **(4) Make Insurance Available and Accessible (Social Inclusion Goals):** Expand micro-insurance, agriculture insurance, and rural branch coverage.
• **(5) Prevent Unfair Market Practices & Ensure Prompt Claim Settlement:** Stop misleading sales, rebating, and unreasonable claim delays.

2. Three Methods of Regulating Insurers:

• **(i) Legislation (Statutory Law):** Enacting comprehensive laws—such as the **Insurance Act, 2079 (2022)** and Insurance Regulations in Nepal.
• **(ii) Regulatory Authority (Administrative Supervision):** Empowering an autonomous regulator—the **Nepal Insurance Authority (NIA / Beema Pradhikaran)**—to issue directives, inspect insurers, and enforce solvency.
• **(iii) Judicial / Court Decisions & Ombudsman Rulings:** Interpreting ambiguous policy clauses and resolving claim disputes.

3. Seven Major Areas of Insurance Regulation & Functions of Nepal Insurance Authority (NIA):

• **1. Licensing & Minimum Paid-Up Capital:** Vetting promoters and enforcing paid-up capital floors (Rs 5 Billion for Life Insurers, Rs 2.5 Billion for Non-Life Insurers, Rs 750 Million for Micro-Insurers, and Rs 10–20 Billion for Reinsurers in Nepal).
• **2. Financial Solvency, Risk-Based Capital (RBC) & Technical Reserves:** Enforcing minimum solvency ratios, unearned premium reserves, catastrophe reserves, and annual actuarial valuations.
• **3. Investment Regulation:** Directing how insurers invest policyholder funds across government bonds, commercial/development bank FDs, infrastructure/hydropower, real estate, and listed shares.
• **4. Rate & Tariff Regulation:** Approving premium rates and setting tariffs for motor third-party, property/fire, and agriculture insurance.
• **5. Policy Form & Product Approval:** Vetting all policy wordings and riders before launch.
• **6. Licensing & Supervision of Market Intermediaries:** Licensing insurance agents, third-party facilitators, and loss surveyors, and enforcing corporate governance and AML/CFT compliance.
• **7. Policyholder Protection & Quasi-Judicial Claim Settlement:** Hearing policyholder complaints against insurers and ordering claim payments with interest when insurers delay unfairly.

SECTION 8: BNK 209 — WORKED ACTUARIAL NUMERICALS, QUESTION BANK & CASE PLAYBOOK

8.1 Complete Step-by-Step Worked Numerical Problems for BNK 209 Exams

Numerical 1: Human Life Value (HLV) Calculation (TU Model Question)

Question: Mr. Suman Thapa, aged 40, is married and has two children. He earns Rs 600,000 annually and plans to retire at age 60. Of this amount, Rs 30,000 is used for taxes, life and health insurance, and his personal needs. The remaining amount is used to support his family. Using a discount rate of 6 percent, calculate the Human Life Value of Mr. Thapa. [PVIFA(6%, 20 yrs) = 11.4699]

Step-by-Step Solution:

• **Step 1: Remaining Working Years (n):** n = Planned Retirement Age - Current Age = 60 - 40 = **20 years**.

• **Step 2: Net Annual Income Contributed to Family:**

Annual Family Share = Gross Annual Earnings - (Taxes + Insurance + Personal Self-Maintenance)

Annual Family Share = Rs 600,000 - Rs 30,000 = **Rs 570,000 per year**.

• **Step 3: Present Value of Family's Share of Earnings (Human Life Value, HLV):**

HLV = Annual Family Share x PVIFA(6%, 20 years)

HLV = Rs 570,000 x 11.4699 = **Rs 6,537,843**.

Numerical 2: Comprehensive HLV, Financial Needs Approach & 1-Year Term Net Single Premium (SDC Preboard 2026)

Question: Shree Sharma (age 35, software engineer, retires at age 60) earns NPR 18,00,000/yr; after taxes and personal living costs, NPR 12,00,000/yr is available for family support. Obligations: Home loan = NPR 50,00,000; Daughter's education fund (in 10 yrs) = NPR 30,00,000; Emergency fund = NPR 10,00,000; Monthly family living expenses = NPR 60,000 (NPR 7,20,000/yr). Discount rate = 7%; Mortality probability at age 35 (q_35) = 0.0020.

Step-by-Step Solution:

• **(a) Human Life Value (HLV) Approach:**

- Working years (n) = 60 - 35 = **25 years**; PVIFA(7%, 25 yrs) = [1 - (1.07)^(-25)] / 0.07 = **11.653583**
- **HLV** = NPR 12,00,000 x 11.653583 = **NPR 1,39,84,300 (NPR 13,984,300)**.

• **(b) Financial Needs Approach:**

- Immediate Home Loan Payoff = **NPR 50,00,000**
- Emergency Fund = **NPR 10,00,000**
- Daughter's Education Fund: PV at 7% for 10 yrs = 30,00,000 / (1.07)^10 = **NPR 15,25,048** (or Nominal = **NPR 30,00,000**)
- PV of Family Living Expenses (NPR 7,20,000/yr for 25 yrs @ 7%) = 7,20,000 x 11.653583 = **NPR 83,90,580**
- **Total Insurance Need** = 50,00,000 + 10,00,000 + 15,25,048 + 83,90,580 = **NPR 1,59,15,628** (or **NPR 1,73,90,580** with nominal education fund).

• **(c) 1-Year Term Net Single Premium (NSP) for Face Value of NPR 10,00,000:**

- Expected Death Claim at Year-End = Face Value x q_35 = NPR 10,00,000 x 0.0020 = **NPR 2,000**
- **Net Single Premium (NSP)** = Expected Claim / (1 + i) = NPR 2,000 / 1.07 = **NPR 1,869.16**.

Numerical 3: Coinsurance (Condition of Average), Deductible & Pro-Rata Contribution

Question: A factory building has a Replacement Cost of Rs 20,000,000 and an estimated physical depreciation of 25%. It is insured under a Fire Insurance Policy carrying an 80% Coinsurance Clause (based on Actual Cash Value) and a Rs 50,000 straight deductible. The owner carries Rs 9,000,000 of fire insurance. A fire causes a partial loss with a replacement cost of Rs 3,000,000.

Step-by-Step Solution:

• **Step 1: Actual Cash Value (ACV) of Building and Loss:**

- ACV of Building = Rs 20,000,000 x (1 - 0.25) = **Rs 15,000,000**
- ACV of Loss = Rs 3,000,000 x (1 - 0.25) = **Rs 2,250,000** (or if loss is already given in ACV = Rs 3,000,000).

• **Step 2: Insurance Required Under 80% Coinsurance:**

- Insurance Required = 80% of Rs 15,000,000 = **Rs 12,000,000** (Owner carried only Rs 9,000,000 -> underinsured!).

• **Step 3: Coinsurance Recovery Before & After Deductible:**
- Coinsurance Ratio = Carried / Required = 9,000,000 / 12,000,000 = **75% (0.75)**
- Recovery on ACV Loss of Rs 2,250,000 = (0.75 x Rs 2,250,000) - Rs 50,000 deductible = Rs 1,687,500 - Rs 50,000 = **Rs 1,637,500**.

8.2 Top 25 Rapid-Fire Group A (2-Mark) Definitions & Answers (BNK 209)

Q#	Exam Brief Question (2 Marks)	Concise Full-Marks Answer Key
1	Define Risk, Peril & Hazard.	Risk is uncertainty concerning the occurrence of a loss; Peril is the direct cause of loss (fire, flood); Hazard is a condition increasing loss frequency or severity.
2	Risk vs. Uncertainty?	Risk has objectively measurable probabilities from historical data and is insurable; Uncertainty is subjective doubt whose probabilities cannot be quantified.
3	List the 4 Types of Hazards.	(1) Physical Hazard, (2) Moral Hazard (dishonesty/fraud), (3) Attitudinal / Morale Hazard (carelessness), and (4) Legal Hazard.
4	What is Systemic Risk?	The risk of collapse of an entire financial system or market due to financial interlinkages, where one major institution's failure triggers a domino effect.
5	What is Enterprise Risk?	A holistic concept encompassing all pure, speculative/financial, operational, cyber, and strategic risks faced by an organization, managed via ERM.
6	Define Risk Pooling.	Spreading unexpected losses incurred by a few individuals over a large group of homogeneous exposure units so that average loss substitutes for actual loss.
7	Objectives of Risk Management?	Pre-Loss: Economy, anxiety reduction, legal compliance. Post-Loss: Survival, continuity of operations, earnings stability, continued growth, social responsibility.
8	What is Loss Forecasting & its 3 Tools?	Predicting future loss frequency and severity using: (1) Probability Analysis, (2) Regression Analysis, and (3) Loss Distributions (Poisson/Lognormal).
9	How does RMIS help Risk Management?	A computerized database that tracks property exposures, policy limits, and claims histories to automate loss forecasting, trend analysis, and renewals.
10	Risk Appetite vs. Risk Tolerance?	Risk Appetite is the broad strategic level of risk a firm is willing to pursue; Risk Tolerance is the specific quantitative maximum acceptable deviation around a target.
11	What is Value at Risk (VaR)?	A statistical metric measuring the maximum expected financial loss on a portfolio or firm over a specified time horizon (e.g., 10 days) at a given confidence level (e.g., 99%).
12	What is Risk Avoidance?	A risk-control technique where a firm refuses to engage in or abandons a high-risk activity, reducing the probability (frequency) of loss to zero (0%).
13	Define Insurance & its 4 Characteristics.	A legal arrangement transferring pure risk to an insurer for a premium. Characteristics: (1) Pooling of losses, (2) Fortuitous losses, (3) Risk transfer, (4) Indemnity.
14	What is Adverse Selection?	The tendency of higher-than-average risk applicants to seek insurance at standard rates more aggressively than low-risk persons, controlled via underwriting.
15	Named-Perils vs. Open-Perils Coverage?	Named-Perils covers only perils explicitly listed in the policy; Open-Perils (All-Risks) covers all accidental causes of loss except those specifically excluded.

Q#	Exam Brief Question (2 Marks)	Concise Full-Marks Answer Key
16	Endorsement vs. Rider?	Both are written amendments modifying a standard policy—called an 'Endorsement' in Property/Casualty insurance and a 'Rider' in Life/Health insurance.
17	Term Life vs. Cash-Value Life?	Term Life offers temporary pure death protection with zero cash value at lowest cost; Cash-Value Life (Whole/Endowment) combines protection with a savings element.
18	Life Insurance Tax Rules in Nepal?	Under Income Tax Act 2058: Annual premiums up to Rs 40,000 are tax-deductible; Death benefits are 100% tax-exempt; Maturity net gain carries 5% final withholding tax.
19	Meaning of 'Fire' in Fire Insurance?	Must involve (1) actual ignition/flame (not mere heating/scorching) and (2) be accidental/fortuitous and a 'Hostile Fire' escaping its normal container.
20	Third-Party Motor Liability Insurance?	Compulsory auto coverage where the insurer (2nd party) pays compensation for bodily injury, death, or property damage caused by the insured's vehicle (1st party) to outsiders (3rd party).
21	What is Professional Liability (E&O;)?	Errors & Omissions / Indemnity insurance protecting professionals (IT consultants, doctors, CAs, engineers) against client lawsuits and defense costs for negligence or mistakes.
22	What is Personal Accident (PA) Insurance?	A benefit policy paying fixed compensation for accidental death, permanent total disability (PTD), permanent partial disability (PPD), or temporary wage loss.
23	What is Commercial Liability Insurance?	Business insurance protecting firms against legal liability for bodily injury, property damage, or defective products/services caused to third parties.
24	Two reasons NIA regulates insurers?	(1) To maintain insurer financial solvency and protect policyholders' funds, and (2) To ensure fair, adequate rates and prompt claim settlement.
25	What is Social Insurance in Nepal?	Compulsory contribution-based protection (e.g., Nepal's SSF: 20% employer + 11% employee = 31% of basic pay) covering Medical/Maternity, Accident, Dependent, and Old-Age Pension.

8.3 Master Guide to Group D Case Studies in BNK 209 (20-Mark Scoring Playbook)

Across all TU Model, Past (April 2026), and SDC Preboard papers, Group D (20 Marks) in BNK 209 tests either:
• **Case Type A: Entrepreneur / Tech Consultancy / Professional Case (e.g., Aarav Sharma [Everest Tech Solutions] or Bikash Koirala [Horizon Digital Advisory]):**
1. **Question (a) — Define & Identify All Pure Risks and Speculative Risks:**
- Always start by defining *Pure Risk* (loss or no loss; insurable) and *Speculative Risk* (loss, no loss, or gain; uninsurable).
- List **Personal Pure Risks** (premature death of founder, permanent disability, critical illness leaving family unable to pay living expenses, children's schooling, and home/car mortgages), **Property Pure Risks** (fire, lightning, earthquake, flood, or theft destroying office servers, networking equipment, and laptops), and **Liability / Operational Pure Risks** (client data breach or system failure from employee configuration error/negligence, business interruption, employee sickness).
- List **Speculative Risks** (stock market investing/short-term trading gains or losses; entrepreneurial startup investment of personal savings into the consultancy).
2. **Question (b) — Recommend Suitable Life Insurance Policy + Justification + Riders:**
- Recommend a **Term Life Insurance Policy** (or a Term + Child Endowment combination).
- *Justification:* Since the entrepreneur has tied up personal savings in the startup and carries heavy home/car loans and family dependencies, Term Life provides the maximum lump-sum death benefit at the lowest premium, preserving cash flow for the business. (And for a CA/investor like Ram Kumar Subedi, cite *"Buy Term and Invest the Difference"*!).

- *Three Riders to Add:* (i) **Critical Illness (CI) Rider**, (ii) **Permanent Total Disability (PTD) & Accidental Death (AD&D;) Rider**, and (iii) **Waiver of Premium Rider**.
3. **Question (c) — Recommend Property / Fire / Professional Liability Insurance:**
- For client data breaches & employee negligence: Recommend **Professional Liability (Errors & Omissions / Cyber Liability) Insurance** (pays both compensatory damages awarded to the client AND legal defense/court costs).
- For office servers, laptops & furniture: Recommend a **Standard Fire and Allied Perils Policy (extended with Earthquake, Flood, and RSMD endorsements) + Electronic Equipment Insurance (EEI) & Burglary Coverage**.
4. **Question (d) — Why Group Health Insurance for Staff is a Risk Management Tool (Not Just an Expense):**
- Explain 4 points: (i) **Controls Human Resource & Operational Continuity Risk** (fast medical treatment reduces employee absenteeism and project delays in a small 5–6 person team), (ii) **Employee Retention & Talent Attraction** (prevents costly turnover of trained IT staff), (iii) **Reduces Attitudinal Hazard & Human Error** (employees free from family medical anxiety make fewer configuration mistakes), and (iv) **Predictable Tax-Deductible Budgeting** (converts unpredictable medical emergencies into a fixed, tax-deductible group premium).

• **Case Type B: Quantitative Family Financial Planning Case (e.g., Shree Sharma):**

- Show clear step-by-step calculations for: (a) **Human Life Value (HLV)**, (b) **Financial Needs Approach Table**, (c) **Policy & Rider Recommendation**, and (d) **1-Year Term Net Single Premium (NSP)**.

8.4 Additional High-Yield Syllabus Topics: Insurance vs. Gambling/Hedging, Cost of Life Insurance & Progress in Nepal

A. Complete Comparison Tables: Insurance vs. Gambling & Insurance vs. Hedging (Unit 3)

Basis of Comparison	Insurance	Gambling	Hedging (Financial Derivatives)
1. Risk Creation vs. Transfer	Transfers an **existing pure risk** that was already present.	Creates a **new speculative risk** that did not exist before the bet.	Transfers an **existing speculative price/market risk** (FX, interest, commodities).
2. Type of Risk Handled	Deals strictly with **Pure Risk** (Loss or No Loss).	Deals with **Speculative Risk** (Gain or Loss).	Deals with **Speculative Price Risk** (two-sided market fluctuations).
3. Insurable Interest	Requires strict legal **insurable interest** in the subject matter.	No insurable interest required beyond the wager.	Hedger holds an underlying spot asset or anticipated transaction.
4. Principle of Indemnity	Restores the insured to pre-loss financial position (never meant for profit).	Winner profits at the loser's expense.	Offsets spot price loss with derivative gain to lock in a target price.
5. Risk Reduction Mechanism	Reduces objective risk via **Risk Pooling & Law of Large Numbers**.	Increases individual wealth variance and risk.	Shifts risk to speculators without pooling homogeneous units.

B. Determining the Cost of Life Insurance & Rate of Return on Savings Component (Unit 5)

When shopping for life insurance (Unit 5 of TU Syllabus), consumers must compare the true cost of competing policies using two methods and evaluate the rate of return on the cash-value savings component:

• **1. Traditional Net Cost (TNC) Method (Ignores Time Value of Money):**
- *Formula:* **Total Premiums Paid (for n years) - Total Dividends Received - Cash Surrender Value at Year n = Total Net Cost** (divided by n years and thousands of face value).
- *Limitation:* Completely ignores the Time Value of Money (interest foregone on premiums paid over 10 or 20 years), often making cash-value policies artificially appear to have a "negative net cost."

• **2. Interest-Adjusted Cost Methods (Surrender Cost Index & Net Payment Cost Index — Incorporates TVM):**

- **Surrender Cost Index (if policy is surrendered at year 10 or 20):**
1. Accumulate annual premiums at an assumed interest rate (e.g., 5% p.a.) using Future Value of Annuity Due [FVAD(5%, 20) = 34.719].
2. Subtract the accumulated Future Value of policy dividends at 5%.
3. Subtract the Cash Surrender Value at the end of year 20.
4. Divide the remainder by FVAD(5%, 20) and per Rs 1,000 of face value to get the true annual interest-adjusted surrender cost per Rs 1,000.
- **Net Payment Cost Index (if policy is held until death without surrendering):** Same steps as above, except *Cash Surrender Value is NOT subtracted*.

• 3. Linton Yield (Rate of Return on the Savings Component of Cash-Value Life Insurance):
Measures the implicit compound annual rate of return earned on the extra premium paid for a Cash-Value policy compared to buying an equivalent pure Term Life policy and investing the premium difference in a separate savings fund ("Buy Term and Invest the Difference").

C. Progress, Challenges & Role of Life and Non-Life Insurance in Nepal (Units 5, 6 & 7):

• 1. Institutional Growth & Consolidation under NIA:
Following the Nepal Insurance Authority's capital hike (Rs 5 Billion for Life Insurers and Rs 2.5 Billion for Non-Life Insurers) and merger wave, Nepal's insurance sector consolidated into **14 Life Insurance Companies, 14 Non-Life Insurance Companies, 7 Micro-Insurance Companies, and 2 Domestic Reinsurance Companies** (Nepal Reinsurance & Himalayan Reinsurance).

• 2. Expansion of Insurance Coverage (Penetration):
Life insurance coverage in Nepal has expanded from less than 10% a decade ago to over **42%–44% of the total population** (including foreign employment term life policies), with insurance premiums contributing approximately 3.5%–4.0% of Nepal's GDP.

• 3. Role of Insurance Industry in Nepal's Financial & Economic Development (10-Mark Exam Question):
1. *Mobilization of Long-Term Contractual Savings:* Collects small recurring premiums from millions of households across all 7 provinces into a massive long-term capital fund (exceeding Rs 650+ Billion).
2. *Backbone of Banking Sector Liquidity & Infrastructure Investment:* Insurers are the largest institutional fixed-deposit holders in Nepalese commercial and development banks and invest heavily in government bonds, hydropower projects, agriculture, and NEPSE securities.
3. *Collateral Security for Bank Credit & Trade:* Enables banks to extend home mortgages, auto loans, industrial project loans, and import Letters of Credit safely.
4. *Protection of Migrant Workers & Rural Farmers:* Mandatory Foreign Employment Term Life Insurance protects over 1.5 million Nepalese migrant workers' families, while 80%-subsidized Agriculture & Livestock Insurance protects rural livelihoods.
5. *Retention of Foreign Currency via Domestic Reinsurance:* Nepal Re and Himalayan Re retain billions of Rupees of reinsurance premiums domestically.
6. *Employment & Fiscal Contribution:* Provides employment to over 12,000 staff and 200,000+ licensed agents while contributing billions in corporate and withholding taxes.